

ABSTRACT CLASSES AND METHODS

1. Abstract methods are those types of methods that don't require implementation for its declaration. These methods don't have a body which means no implementation.

2. Syntax:

```
abstract void career(String name);
```

3. If a class contains one or more abstract methods, the class must also be declared as abstract.

4. Abstract methods must be overridden in subclasses.

5. You cannot create object of abstract class.

But you can use it as reference variable. For example: `Parent mom = new Daughter();`

If you want to create object of abstract class, you need to define abstract methods at the time of constructing objects. For example:

objects. For example:

```
Parent mom = new Parent() {  
    @Override  
    void career() {  
  
    }  
}
```

Reason: If you create object of abstract class and call abstract method, it will be unable to do so because abstract method don't have body.

6. You cannot create abstract constructors.

Although you cannot create object of abstract class, you can create its constructor because it can be called through subclass constructor using "super".

7. You cannot create "abstract static" methods because abstract methods need to be overridden and static methods can't be overridden.

8. You can create abstract methods as well as static methods in abstract classes.

9. "Abstract" class cannot be "final" because "final" class cannot be inherited but "Abstract" class need to be overridden in "subclass".

Do's and Dont's in Abstract classes and methods

Do's Dont's

- | | |
|-----------------------|---|
| 1. Abstract methods | 1. Abstract constructor |
| 2. Static methods | 2. Abstract Static methods |
| 3. Non-static methods | 3. Cannot create object (Some conditions) |
| 4. Cannot be final | |